

CFNet: Optimizing Remote Sensing Change Detection Through Content-Aware Enhancement

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Abstract—Change detection is a crucial and widely applied task in remote sensing, aimed at identifying and analyzing changes occurring in the same geographical area over time. Due to variability in acquisition conditions, bitemporal remote sensing images often exhibit significant differences in image style. Even with the powerful generalization capabilities of DNNs, these unpredictable style variations between bitemporal images inevitably affect the model’s ability to accurately detect changed areas. To address issue above, we propose the Content Focuser Network (CFNet), which takes content-aware strategy as a key insight. CFNet employs EfficientNet-B5 as the backbone for feature extraction. To enhance the model’s focus on the content features of images while mitigating the misleading effects of style features, we develop a constraint strategy that prioritizes the content features of bitemporal images, termed Content-Aware. Furthermore, to enable the model to flexibly focus on changed and unchanged areas according to the requirements of different stages, we design a reweighting module based on the cosine distance between bitemporal image features, termed Focuser. CFNet achieve outstanding performance across three well-known change detection datasets: CLCD (F1: 81.41%, IoU: 68.65%), LEVIR-CD (F1: 92.18%, IoU: 85.49%), and SYSU-CD (F1: 82.89%, IoU: 70.78%).

Index Terms—Change detection, content-aware, feature focuser.

I. INTRODUCTION

REMOTE sensing change detection (RSCD) is the process of identifying and quantifying changes in an object, phenomenon, or landscape by comparing images acquired through remote sensing technology at different times. This technique is crucial for understanding changes in land cover, urban growth, environmental shifts, and natural disasters [1]. The process involves comparing bitemporal or multitemporal satellite imagery or aerial photos, often with the help of advanced algorithms, to detect changes such as deforestation, urban expansion, or shoreline shifts [2][3]. In addition, RSCD encompasses various

tasks, including bitemporal binary change detection, bitemporal multiclass change detection, and multitemporal change analysis. In this study, we focus specifically on bitemporal binary change detection, where the goal is to distinguish changed and unchanged areas between two input images.

Advancements in remote sensing technology have led to a rapid increase in the diversity of platforms, ranging from spaceborne (e.g., satellites, spacecraft) to airborne (e.g., drones, balloons) and ground-based platforms (e.g., sensing towers, vehicles) [4][5]. This diversity has significantly increased the complexity of image style features. Even within the same platform, variations in atmospheric conditions, lighting, sensor calibration, and platform trajectory can introduce discrepancies between bitemporal images [6],[7].

While differences in style features are often emphasized in heterogeneous images (e.g., optical versus SAR), similar variations in style can also occur within homogeneous multispectral images. These discrepancies in style can complicate the change detection process even when the images are from the same platform. Such style variations can exacerbate the challenge of isolating meaningful content changes, as they introduce unnecessary interference that confounds the model’s ability to focus on the actual changes in content. In particular, when the style differences between two images are complex and unpredictable, the model may misinterpret unchanged areas—where the only difference lies in style—as changed areas. This misclassification can lead to erroneous results, which undermines the accuracy of the change detection task. Therefore, addressing the interference caused by these inherent style discrepancies is crucial for ensuring that the model focuses solely on the true content changes, which is the primary goal of change detection.

Specifically, the definition of “content” in this article is based on self-similarity, highlighting that human perception identifies objects by their appearance relative to their surroundings rather than their absolute appearance [8], [9]. Our definition of “style” in this article is a distribution over features extracted by a deep neural network, which is mainly caused by imaging conditions including environmental conditions, sensor parameters, acquisition settings, etc [10].

In recent years, Siamese neural networks have dominated RSCD by leveraging shared weights for bitemporal feature extraction [2], [11], [12]. However, disentangling content and style remains a challenge due to style discrepancies from varying imaging conditions (e.g., lighting, atmosphere, sensor calibration). These differences can mislead models, hindering their focus on actual changes. To address this, CiDL employs dual

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The code and pretrained models of CFNet are publicly released at <https://github.com/wifiBlack/CFNet>.

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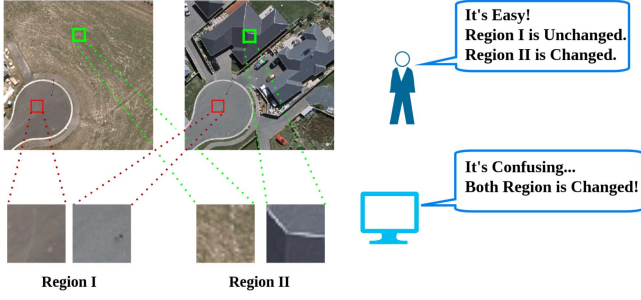


Fig. 1. Humans can easily identify the changed content areas between two images taken under different conditions, without being significantly influenced by image style factors such as brightness, contrast, etc. However, this task is sometimes difficult for computer.

Y-shaped networks with cross-domain translation to suppress style-induced noise [13], while CCNet introduces a multiresolution parallel structure and an auxiliary image restoration task to enhance content-style separation [10]. Yet, these methods overlook the structural consistency of unchanged areas, a key cue for accurate detection. Moreover, the lack of mechanisms to dynamically balance attention between changed and unchanged areas limits their adaptability in complex scenarios.

As shown in Fig. 1, humans can easily identify the content changed areas between two images taken under different conditions, without being significantly influenced by image style factors such as brightness and contrast. This is because when determining whether an area has changed, humans tend to focus more on the internal structure of the area rather than its style features. In fact, this aligns with the cognitive mechanisms of human perception, where humans are accustomed to paying attention to the internal structural information of an object. Based on the above ideas, we propose that even when complex style differences exist between two bitemporal remote sensing images, the difference in feature vectors at any two locations within unchanged areas remains similar. Therefore, the differences within the set of feature vector differences across any two positions in the unchanged areas should be as small as possible. This set can represent the internal structural characteristics of unchanged areas, as described earlier. Similarly, in the changed areas, the differences within the set of feature vector differences across any two positions in the changed areas should be as large as possible.

As illustrated in Fig. 2, Image T1 and Image T2 exhibit significant style differences, and thus, even in the unchanged areas of the two images, the distribution of features in the feature space is inconsistent. To simplify the representation, we use a two-dimensional Cartesian coordinate system to illustrate the distribution of multidimensional feature vectors in feature space. In the figure, each ellipse represents the distribution of a pixel's features in feature space. We use θ to denote the difference between the features of two sampled points in feature space. Next, we randomly sample two points each from both the changed and unchanged areas. The red boxes in the figure represent sampling points from the unchanged areas, while the green boxes represent those from the changed areas. In the unchanged areas, where the internal structure is similar, the value

of θ_1 is close to the value of θ_2 . In contrast, in the changed areas, where the internal structure varies significantly, the value of θ_3 diverges from the value of θ_4 . By modeling these characteristics, we aim to enhance the model's ability to learn content features in both changed and unchanged areas. In addition, the difference in content features between changed and unchanged areas serves as a mutual constraint, further improving the model's accuracy.

In existing change detection algorithms, the concatenation method is widely used for feature fusion of bitemporal remote sensing images. The reason this method achieves relatively good performance is that DNN have strong generalization capabilities, allowing them to learn the fitting relationship between fused features and labels [2]. However, concatenation does not directly compute the changed areas between two images. Therefore, how to make the model focus more on the changed areas during feature fusion, in order to better align with the labels, remains a crucial challenge. In addition, existing change detection algorithms often pay much more attention to the model's ability to fit the changed areas in the labels. However, the binary classification nature of bitemporal binary change detection tasks means that the accuracy of fitting unchanged areas also significantly impacts the detection results for changed areas in the final prediction map. Therefore, effectively leveraging the mutual constraints between changed and unchanged areas to further enhance the robustness of the model remains a pressing challenge. In this article, we creatively designed a plug-and-play focuser module that allows the model to flexibly focus on both changed and unchanged areas depending on the task's requirements at different stages. Building on this, our designed Focuser module not only enables the model to explicitly focus on the changed areas during feature fusion but also leverages the mutual constraints between changed and unchanged areas to impose enhanced parameter regularization, thereby improving the model's robustness and accuracy.

In summary, the main contributions are as follows:

- 1) We propose a content-aware strategy, a novel content-based constraint learning strategy that enhances the model's focus on intrinsic content features while reducing the impact of style variations, thereby improving the accuracy and robustness of bitemporal change detection in remote sensing imagery.
- 2) We introduce a plug-and-play focuser module, a novel mechanism that dynamically reweights features to focus on both changed and unchanged areas, leveraging their mutual constraints to enhance parameter regularization and improve model accuracy.

II. RELATED WORKS

A. Deep Learning in Change Detection

Traditional change detection methods, including random forests [14], support vector machines [15], Markov random fields [16], decision tree [17], and conditional random fields [18], [19], rely heavily on handcrafted features and struggle with complex environmental variations. In contrast, deep learning approaches have demonstrated superior performance by automatically extracting hierarchical feature representations,

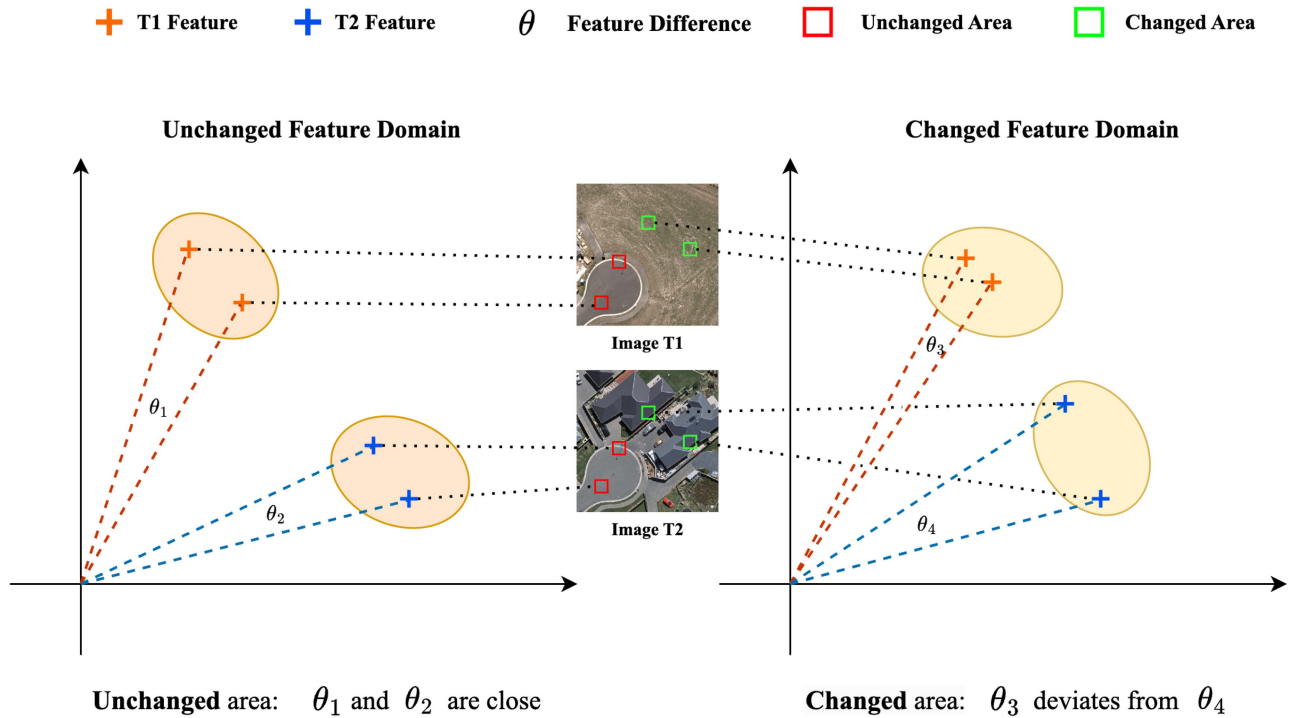


Fig. 2. Image T1 and Image T2 exhibit significant style differences. Each ellipse in figure represents the distribution of a pixel's features in feature space. And we use θ to denote the difference between the features of two sampled points in feature space. Next, we randomly sample two points each from both the changed and unchanged areas. The red boxes in the figure represent sampling points from the unchanged areas, while the green boxes represent those from the changed areas. In the unchanged areas, where the internal structure is similar, the value of θ_1 is close to the value of θ_2 . In contrast, in the changed areas, where the internal structure varies significantly, the value of θ_3 deviates from the value of θ_4 .

effectively addressing challenges such as image quality variations, noise, registration errors, and spatial heterogeneity [20].

Convolutional neural networks (CNNs) have demonstrated remarkable performance in change detection tasks, primarily due to their powerful feature extraction capabilities and nonlinear modeling capacity. Unlike traditional methods, CNNs can automatically learn hierarchical spatial representations, thereby improving change localization and classification accuracy. For instance, Alcantarilla et al. [21] developed a structural change detection system for street-view videos using fully convolutional networks. Shao et al. [22] introduced a dual-channel network incorporating edge information to address challenges in heterogeneous satellite and UAV image change detection. Lin et al. [23] proposed P2V-CD, a framework that constructs pseudovideo sequences and employs decoupled encoders to enhance temporal information processing, significantly improving detection accuracy. More recently, Han et al. [24] introduced CGNet, which integrates change guidance maps and a self-attention module to refine feature representations, improving edge integrity and reducing internal noise in change maps. In addition, Dong et al. [25] proposed EfficientCD, which utilizes EfficientNet as its backbone and integrates ChangeFPN to enhance multiscale feature aggregation through progressive upsampling, achieving state-of-the-art performance.

Transformer-based approaches have recently gained significant attention in change detection due to their capability to model long-range dependencies and capture spatial-temporal correlations more effectively than CNNs. Chen et al. [26] proposed the

bitemporal image transformer (BIT), which formulates change detection as a token-based representation learning problem. By encoding bitemporal images into a compact set of semantic tokens and leveraging a transformer encoder-decoder framework, BIT efficiently models spatial-temporal dependencies while significantly reducing computational costs. Bandara et al. [27] introduced ChangeFormer, a Siamese network architecture that integrates a hierarchically structured transformer encoder with an MLP decoder, effectively capturing multiscale contextual information for improved change detection accuracy. More recently, Dong et al. [28] proposed ChangeCLIP, which extends the transformer paradigm by incorporating vision-language pre-training, leveraging the semantic representations of image-text pairs to enhance the robustness of change detection. These advancements demonstrate the increasing impact of transformer architectures in RSCD, further expanding the scope beyond traditional CNN-based frameworks.

Given the complementary strengths of CNNs in capturing local spatial features and transformers in modeling long-range dependencies, hybrid CNN-Transformer architectures have been increasingly explored for change detection. These approaches typically utilize CNNs for low-level feature extraction while leveraging transformers to enhance global contextual representation. Yuan et al. [29] proposed STTransUNet, which integrates a UNet-based CNN for hierarchical feature extraction with a transformer module to capture global dependencies, along with a cross-enhanced adaptive fusion module to refine bitemporal feature representations. Xu et al. [30] introduced HATNet, a

hybrid attention-aware transformer network that incorporates self-attention and coordinate-attention mechanisms to improve multiscale feature extraction and alignment, ensuring better spatial coherence in change maps. These hybrid models effectively combine the advantages of CNNs and transformers, demonstrating superior performance in RSCD by balancing local detail preservation and global feature interaction.

B. Separation of Content and Style

Style transfer refers to the creation of artistic imagery by separating and recombining the content and style of images. Recently, image style transfer has garnered significant attention. In this field, some researchers have proposed methods to effectively separate image style from image content, achieving impressive results.

Gatys et al. [31] illustrated the capability of CNNs to separate and recombine image content and style, pioneering the neural style transfer (NST) technique. This method, which applies CNNs to render content images in diverse styles, has since gained significant traction in both academia and industry, inspiring numerous approaches to enhance or expand upon the original NST framework [32].

Xu et al. [33] introduced a coanalysis method which enables efficient style transfer and synthesis of new 3-D objects by defining a correspondence-free style signature for clustering, facilitating part-level correspondence within style clusters. Zhange et al. [34] proposed generalized style transfer network enables effective style-content separation using conditional dependence, allowing for versatile style transfer and generalizability across new styles and contents. Zhang et al. [35] proposed framework that enables a single style transfer model to generalize across multiple styles and contents by leveraging separate style and content encoders, enhancing its applicability to unseen styles and contents. In 2021, StyleMix and StyleCutMix improve robustness by selectively mixing content and style information in data augmentation, enhancing model generalization and resilience against adversarial attacks [36].

Some researchers have creatively identified that the strategy of separating image content and style can be effectively applied to RSCD tasks. Fang et al. [13] first introduced the concept of content-style separation into RSCD tasks. Their proposed CiDL integrates a dual learning algorithm with disentangled representation theory to separate content and style features, suppressing style discrepancies in unchanged areas while highlighting content changes, thereby improving accuracy and reducing dependency on labeled data. Building on this foundation, Cheng et al. [10] further advanced disentangled representation learning by refining the separation of content and style into distinct subspaces, effectively addressing pseudochanges caused by varying imaging conditions and platforms, and enhancing the robustness of change detection frameworks.

In this article, we extend the concept of content-style separation to the complex task of change detection in remote sensing, with a novel design tailored specifically to enhance the model's ability to focus on the content features of bitemporal remote sensing images. This approach mitigates the risk of the model

TABLE I
DETAILED ARCHITECTURE OF ENCODER

Stage	Operator	Kernel size	Resolution	Channels	Layers
1	Conv	3 x 3	128 x 128	48	1
2	MBConv1	3 x 3	128 x 128	24	3
3	MBConv6	3 x 3	64 x 64	40	5
4	MBConv6	5 x 5	32 x 32	64	5
5	MBConv6	3 x 3	16 x 16	128	7

being misled by complex style differences between the two temporal images during training.

III. METHOD

A. Overall Architecture

In this article, we propose a novel content-based constraint learning strategy, content-aware, which focuses on content features. In addition, we have ingeniously designed a plug-and-play focuser module that enables the model to flexibly focus on changed and unchanged areas.

As illustrated in Fig. 3, the complete architecture is divided into four principle stages: feature extraction, content focuser decoder, change decoder, and loss computation. In the Stage I, feature extraction, we employ a part of the EfficientNet-B5 backbone as the encoder to extract rich multiscale features from the bitemporal images [37]. In Table I, the term MBConv refers to a specific module within the architecture. The number following MBConv (e.g., 1 or 6) indicates the expansion factor, which determines how many times the number of input feature matrix channels will be expanded by the first 1×1 convolution layer within the MBConv module. The kernel size represents the size of the convolutional kernel used in the Depthwise convolution, a critical operation within the MBConv module [38]. The "resolution" column specifies the input resolution for each stage, while "channels" refers to the number of output feature matrix channels produced after passing through the stage. Finally, "layers" indicates the number of times the operator structure is repeated in a given stage. It is worth noting that the bitemporal images share the same weights during the encoding process. The detailed structure of encoder is presented in Table I. The feature maps extracted from stage 2 to stage 5 of the bitemporal images are used as the output of the encoder, denoted as $F_{a1}, F_{a2}, F_{a3}, F_{a4}$ for image T1, and $F_{b1}, F_{b2}, F_{b3}, F_{b4}$ for the image T2, respectively.

In the Stage II, the content focuser decoder, the model first decodes the multiscale features extracted from the bitemporal images to obtain content features at different scales, as shown in Fig. 4.

It is important to clarify that in Fig. 4, F_i , $i = 1, 2, 3, 4$ (i.e., F_{ai} or F_{bi}), denotes the output of the encoder, which serves as the input to the content decoder. The content decoder then processes these encoder features to generate the corresponding content features C_i (i.e., C_{ai} or C_{bi}). Although the two change decoder modules share the same architecture, they operate with independent weights. The Agg module is used to aggregate the feature maps from adjacent scales of the encoder's output.

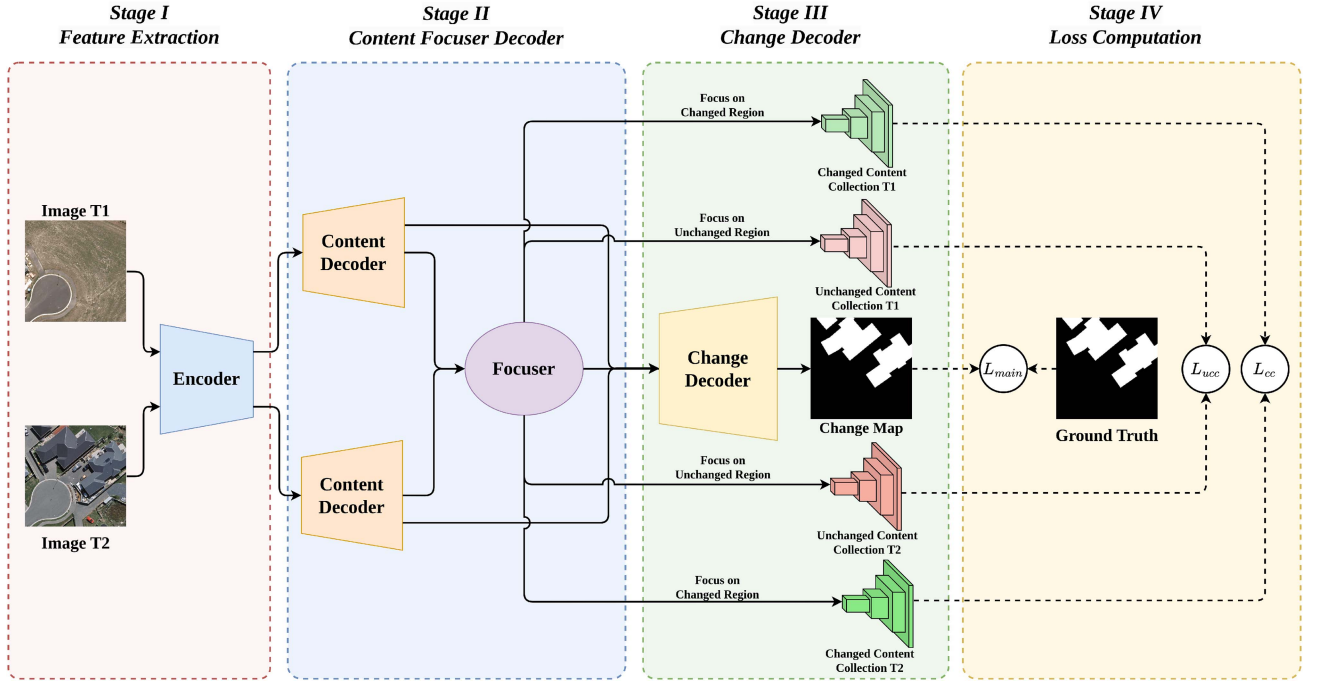


Fig. 3. Overall architecture of CFNet. The architecture is divided into four key stages: feature extraction, content focuser decoder, change decoder, and loss computation. In Stage I, a partial EfficientNet-B5 backbone extracts multiscale features from bitemporal images. In Stage II, The decoder extracts content features, and the focuser module generates reweighting maps to separate changed and unchanged content. In Stage III, content features and reweighting maps are leveraged to generate the change map. In Stage IV, the total loss consists of the main loss L_{main} , computed using MSE loss between the change map and the ground truth, and the auxiliary losses L_{ucc} and L_{cc} , which work collaboratively to distinguish changed and unchanged areas, further enhancing the model's performance. L_{cc} denotes “changed content loss” and L_{ucc} denotes “unchanged content loss.”

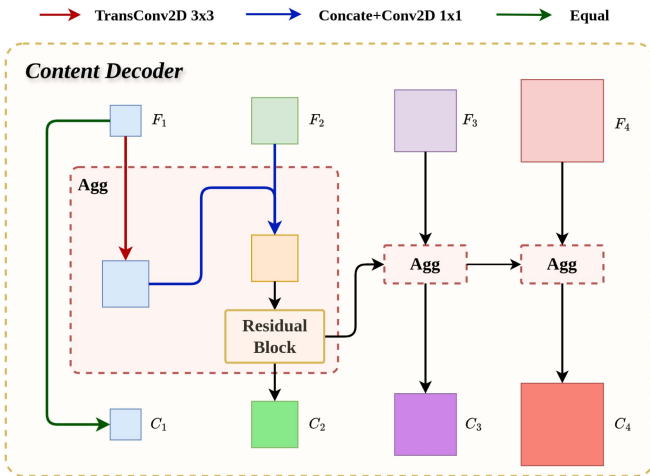


Fig. 4. Detailed architecture of content decoder. $F_i, i = 1, 2, 3, 4$ denotes the output of the Encoder, specifically F_{ai} or F_{bi} . $C_i, i = 1, 2, 3, 4$ represents content feature, specifically C_{ai} or C_{bi} . The Agg module is used to aggregate the feature maps from adjacent scales of the encoder's output.

Specifically, it first upsamples the lower resolution encoder features using transposed convolutions to match the spatial scale of the higher resolution features. These features are then concatenated along the channel dimension, followed by a convolutional layer that fuses the combined features. A residual block is further applied to enhance the integration of semantic

information, ultimately producing the aggregated content features. The multiscale content features are subsequently passed through the focuser module, which generates reweighting maps corresponding to each scale. These reweighting maps, denoted as $RM_i, i = 1, 2, 3, 4$, retain the same scale as their respective input content features, ensuring alignment across scales. Subsequently, the focuser module reweights the changed and unchanged areas of the bitemporal images, effectively separating changed content features from unchanged content features. This results in the “changed content collection” and “unchanged content collection” for the bitemporal images. The structure and function of the focuser module will be discussed in detail in Section III-B.

In the Stage III, change decoder, the content features at each scale are fully fused, and the reweighting maps generated in the Stage II are utilized at each decoding stage to focus on the changed areas, eventually producing the change map. The detailed structure of change decoder is illustrated in Fig. 5. In this process, the content features from bitemporal images at each scale are first thoroughly integrated within the Fuser module using 3-D convolution and feature concatenation. These fused features are then progressively merged across adjacent layers through the designed Agg modules, ultimately producing the final Change Map. In addition, the specific structure of the Agg module is detailed in Fig. 4.

In the Stage IV, loss computation, the change map is compared with the ground truth to compute the L_{main} , which is calculated using MSE loss. At the same time, “changed content collection”

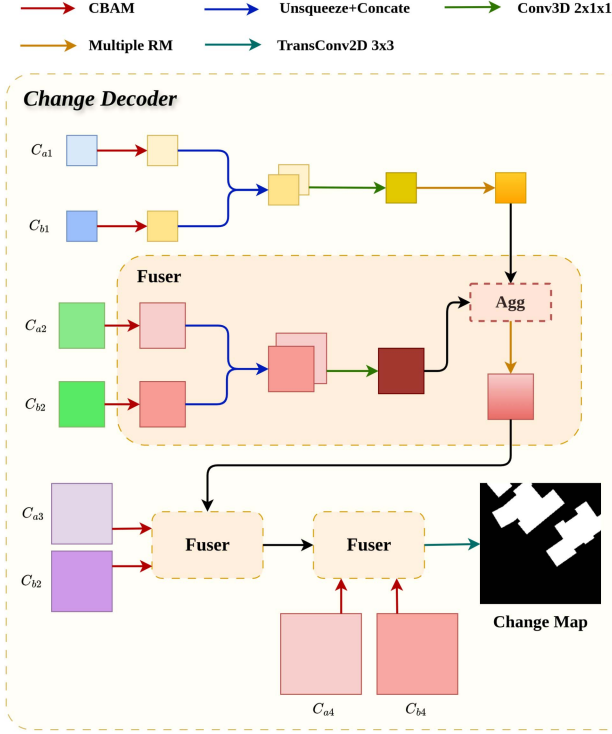


Fig. 5. Detailed architecture of change decoder. The red arrows labeled as “CBAM” represents convolutional block attention module [39]. It is a lightweight module that enhances feature representation by applying channel and spatial attention, helping the network focus on the most relevant features and areas. The blue arrows labeled “Unsqueeze+Concat” indicates that two inputs are each expanded by an identical new dimension, concatenated along this new dimension to produce an output, which is then used for subsequent 3-D convolution operations. The yellow arrows labeled as “multiple RM” represents performing a dot product between the feature map at the starting point of the arrow and the corresponding scale RM_i from focuser module.

and “unchanged content collection” from the Stage II leveraged for self-supervision, enabling the computation of the “changed content loss” and “unchanged content loss.” This computation is central to the content-aware strategy, which will be discussed in detail in Section III-C.

B. Focuser Module

In previous research, researchers have typically chosen to concatenate feature maps at multiple scales and then leverage the large number of model parameters to fit the relationships with changed areas. However, this approach, to some extent, overlooks the essence of the change detection task. We argue that the essence of bitemporal binary change detection is a binary classification task with two inputs. More specifically, the task involves predicting whether a pixel belongs to a changed area or an unchanged area based on bitemporal remote sensing images. Therefore, we believe the model should flexibly focus on features within the changed area and unchanged area separately during the decoding stage, thereby reducing the reliance on strong model fitting ability. This approach also enables the model to learn features from the changed area and unchanged area independently. Based with this idea, we propose a plug-and-play

focuser module, with its detailed structure shown in Fig. 6. It is important to note that the focuser module performs identical computations on the content features of bitemporal remote sensing images at different scales, outputted by the content decoder. Therefore, in the figure, we use C_{ai} and C_{bi} to represent the content features of bitemporal images at the same scale, illustrating the consistent computation process of the focuser module across all scales.

As shown in Fig. 6, C_{ai} and C_{bi} represent the content features of bitemporal images T1 and T2 at a certain scale, as output by the content decoder module. First, we calculate the cosine distance between C_{ai} and C_{bi} at corresponding positions to quantify the degree of content feature differences at each location. The reason for using cosine distance here is that it helps mitigate the influence of absolute feature magnitude differences, thereby reducing the interference from style variations, which allows us to focus more on the content differences. Next, we apply the Tanh function to normalize the cosine distance map between C_{ai} and C_{bi} , generating the reweighting map RM_i for this scale. RM_i reflects the distribution of changed and unchanged areas, where pixels with values closer to 1 are more likely to belong to changed areas, and pixels with values closer to 0 are more likely to belong to unchanged areas. Consequently, in $1 - RM_i$, pixels with values closer to 1 are more likely to be unchanged areas, while values closer to 0 indicate changed areas. Finally, we multiply C_{ai} and C_{bi} by RM_i and $1 - RM_i$, respectively, to obtain CC_{ai} , CC_{bi} , UCC_{ai} , and UCC_{bi} , where “CC” denotes change content features and “UCC” denotes unchanged content features. This approach enables the model to focus on learning both changed and unchanged areas separately during training, thereby improving its performance in change detection task. The computational process mentioned above is presented in the following formulas:

$$\text{Cosine Similarity} = \frac{C_{ai} \cdot C_{bi}}{\|C_{ai}\| \|C_{bi}\|} \quad (1)$$

$$\text{Cosine Distance Map} = \sum_{\text{Channel}} 1 - \text{Cosine Similarity} \quad (2)$$

$$RM_i = \tanh(\text{Cosine Distance Map}) \quad (3)$$

$$CC_{ai} = RM_i \cdot C_{ai} \quad (4)$$

$$CC_{bi} = RM_i \cdot C_{bi} \quad (5)$$

$$UCC_{ai} = (1 - RM_i) \cdot C_{ai} \quad (6)$$

$$UCC_{bi} = (1 - RM_i) \cdot C_{bi} \quad (7)$$

C. Content-Aware Strategy

Due to varying imaging conditions, there are often complex and unpredictable style differences between bitemporal remote sensing images. In change detection task, the primary focus should be on meaningful content changes, but discrepancies in image style often mislead the model’s judgment. We aim to guide the model to learn content features from both changed and unchanged areas through an auxiliary loss function, thereby constraining the network’s fitting.

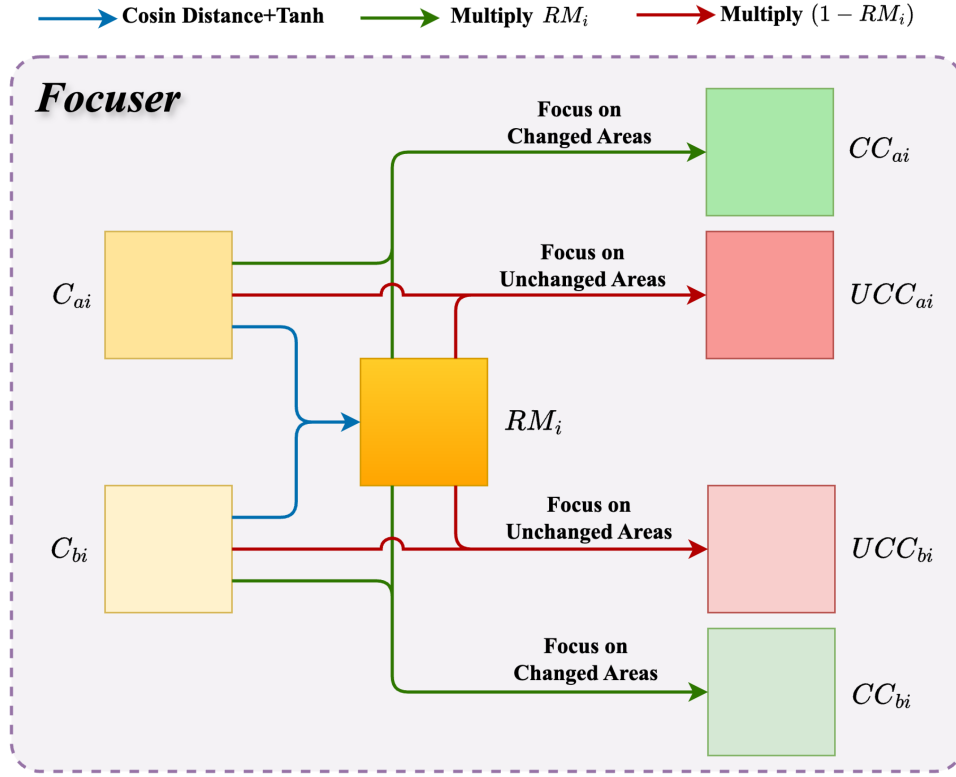


Fig. 6. Detailed architecture of focuser module. C_{ai} and C_{bi} represent the content features of bitemporal images T1 and T2 at a certain scale, as output by the content decoder module. RM_i reflects the distribution of changed and unchanged areas, where pixels with values closer to 1 are more likely to belong to changed areas, and pixels with values closer to 0 are more likely to belong to unchanged areas. Moreover, “CC” denotes changed content features and “UCC” denotes unchanged content features.

As mentioned in the introduction, when humans assess whether a specific area in an image has changed, they can easily discount the interference caused by style differences. This is because human perception of an object is not solely dependent on individual pixels, but rather on the internal structural features of the object. Inspired by self-similarity, the internal structural features referenced here can be mathematically represented by the set of cosine similarities between feature vectors at any two locations within an area. The use of cosine similarity is motivated by its ability to reduce the influence of feature vector magnitude, thereby better mitigating the interference from style differences.

Building on the aforementioned considerations, we propose a novel content-based constraint strategy focused on image content features, named content-aware. We compute the “changed content loss” and “unchanged content loss” at each scale using the “changed content collection” and “unchanged content collection” for both Image T1 and Image T2. As a key component of the content-aware strategy, the detailed computation process is illustrated in Fig. 7. Since the calculation steps for “changed content loss” and “unchanged content loss” differ only in the final step, we explain the computation of “changed content loss” first, followed by the specific details of the final step for “unchanged content loss.”

To ensure the computational efficiency of the model, we performed n rounds of random sampling. In each sampling round, we randomly selected the same two locations from CC_{ai} and

CC_{bi} . The two sets of sampled point pairs were then added as new elements to point pair Set_{ai} and point pair Set_{bi} , respectively. As a result, both point pair Set_{ai} and point pair Set_{bi} ultimately contained n pairs of sampled points. Subsequently, we calculated the cosine similarity of the feature vectors corresponding to each pair of sampled points in point pair Set_{ai} and point pair Set_{bi} , yielding two $1 \times n$ matrices, internal structural Similarity $_{ai}$ and internal structural similarity $_{bi}$. In the formulas, we denote internal structural similarity as “ISS.” These matrices represent subsets of internal structural features, each containing n elements. In addition, to balance computational efficiency and accuracy, in this experiment, we set the number of sampled point pairs n to be the square root of the scale of CC_{ai} and CC_{bi} . For example, if the scale of CC_{ai} is 256×256 , then n is set to 256. Since the sampling process is random in each training iteration, we consider these subsets to approximate the overall internal structural features. In essence, the “changed content loss” should diverge as far from zero as possible, and it is computed according to formula (8). In contrast, the “unchanged content loss” should approach zero, and it is computed using formula (9). Finally, L_{cc} is computed according to formula (10), while L_{ucc} is calculated as formula (11). In addition, it is important to note that to minimize the disparity between L_{main} , L_{cc} , and L_{ucc} , thereby preventing any single loss function from dominating the model’s learning process, we applied appropriate weighting to L_{main} , L_{cc} , and L_{ucc} . The final loss computation is presented in formula (12).

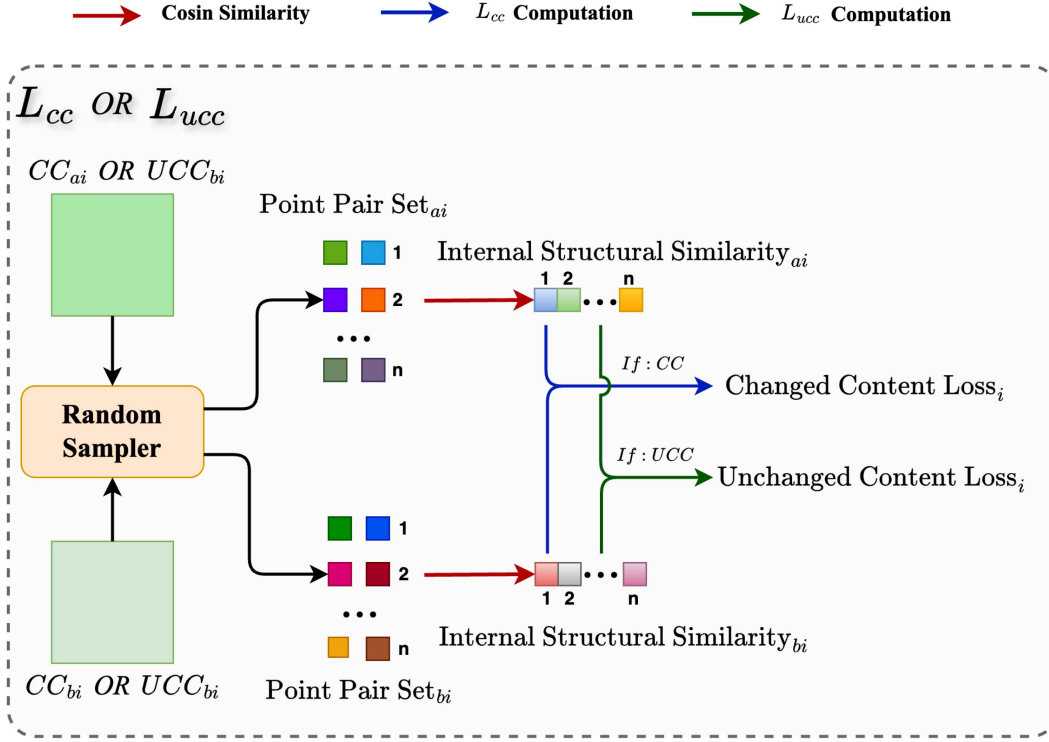


Fig. 7. Detailed calculation process of L_{cc} and L_{ucc} . We leverage the random sampler to randomly sample n different points from the corresponding positions in CC_{ai} , CC_{bi} or UCC_{ai} , UCC_{bi} , resulting in the point pair Set $_{ai}$ and point pair Set $_{bi}$. For each of these sets, we compute the pairwise cosine similarity between the points, yielding $1 \times n$ matrices internal structural Similarity $_{ai}$ and internal structural Similarity $_{bi}$. If the input consists of CC_{ai} and CC_{bi} , we calculate changed content Loss $_i$. Conversely, if the input consists of UCC_{ai} and UCC_{bi} , we compute unchanged content Loss $_i$.

During the experimental process, we set α to 1, β to 0.1, and γ to 0.1

$$\text{Changed Content Loss}_i = 1 - \frac{1}{n} \sum |ISS_{ai} - ISS_{bi}| \quad (8)$$

$$\text{Unchanged Content Loss}_i = \frac{1}{n} \sum |ISS_{ai} - ISS_{bi}| \quad (9)$$

$$L_{cc} = \frac{1}{4} \sum_{i=1}^4 \text{Changed Content Loss}_i \quad (10)$$

$$L_{ucc} = \frac{1}{4} \sum_{i=1}^4 \text{Unchanged Content Loss}_i \quad (11)$$

$$\text{Loss} = \alpha L_{\text{main}} + \beta L_{cc} + \gamma L_{ucc}. \quad (12)$$

IV. EXPERIMENTS

A. Datasets

To validate the effectiveness and robustness of the proposed CFNet, we conducted experiments on three publicly available and well-known RSCD datasets: CLCD [40], LEVIR-CD [41], and SYSU-CD [42]. The CLCD dataset is selected for its agriculture-related change detection focus and the nonfixed style differences between bitemporal images, which allow for evaluating the model's robustness under complex style

variations. The LEVIR-CD dataset is chosen for its suitability in building change detection tasks, particularly for evaluating performance on very high-resolution (VHR) images and analyzing fine-grained building additions and demolitions. The SYSU-CD dataset is used to assess the adaptability of methods across various scenarios, given its focus on localized urban change detection and microlevel urban structural changes. It is worth noting that, due to the complexity of imaging conditions, all three datasets feature bitemporal images with unquantifiable and unpredictable style differences. These inherent style variations present an additional challenge and provide a more comprehensive test for the robustness of the proposed CFNet.

CLCD: The CLCD dataset contains 600 pairs of cropland change samples, divided into 360 pairs for training, 120 pairs for validation, and 120 pairs for testing. The bitemporal images in the CLCD dataset were captured by the Gaofen-2 satellite in Guangdong Province, China, in 2017 and 2019, respectively, with spatial resolutions ranging from 0.5 to 2 m. Each sample pair includes two 512×512 images along with a corresponding binary cropland change label. During training phase, each 512×512 image is divided into four 256×256 patches without overlapping. For testing, we use the original 512×512 images from the test set in the dataset.

LEVIR-CD: LEVIR-CD comprises 637 pairs of VHR (0.5 m/pixel) Google Earth image patches, each sized 1024×1024 pixels. These bitemporal images span a period of 5 to 14 years and capture significant land-use changes, particularly in construction growth. The dataset includes a wide

variety of building types, such as villas, tall apartment buildings, small garages, and large warehouses. The primary focus is on building-related changes, including building growth (e.g., transitions from soil, grass, or construction sites to fully developed structures) and building decline. All bitemporal images are annotated by remote sensing experts with binary labels (1 for the changed, 0 for the unchanged). To ensure annotation quality, each sample is first labeled by one annotator and then reviewed by a second annotator. The fully annotated dataset includes 31 333 individual instances of building changes. During training phase, each 1024×1024 image is divided into multiple 256×256 patches, with a 64-pixel overlap between adjacent patches. Using this method, a single 1024×1024 image is cut into 25 patches of size 256×256 . As a result, we generate 11 125 image pairs for the training set and 1600 image pairs for the validation set. For testing, we use the original 128 image pairs with a size of 1024×1024 from the dataset.

SYSU-CD: The SYSU-CD dataset consists of 20 000 pairs of high-resolution aerial images, each with a resolution of 0.5 m and dimensions of 256×256 pixels. Captured in Hong Kong between 2007 and 2014, this dataset represents a diverse array of urban changes, including the construction of new buildings, suburban sprawl, groundwork prior to construction, alterations in vegetation, road expansions, and offshore construction projects. The dataset is systematically divided into three subsets: the training set containing 12 000 image pairs, the validation set with 4000 pairs, and the test set also comprising 4000 pairs. This organization adheres to widely accepted experimental protocols, making SYSU-CD a valuable resource for assessing and comparing the performance of change detection algorithms across various urban scenarios. Each image pair is annotated with binary labels, indicating whether the pixels have changed or remained unchanged, further enhancing its utility for research in change detection.

B. Benchmark Methods

To demonstrate the superiority of the proposed CFNet, we selected state-of-the-art algorithms for RSCD and conducted a comparative evaluation of CFNet across three datasets: CLCD, LEVIR-CD, and SYSU-CD. These algorithms are categorized into three groups based on their architectural paradigms: CNN-based methods, which leverage convolutional operations for feature extraction and spatial pattern recognition; transformer-based methods, which model long-range dependencies through self-attention mechanisms; and hybrid CNN-transformer methods, which combine the strengths of both architectures to enhance spatial and contextual feature representations.

CNN-based: CDNet [21], SNUNet [43], DDCNN [44], STANet [41], P2V [23], HCGMNet [45], CGNet [24], AF3DNet [46], ChangeEx [47], EfficientCD [25], CDNeXt [48].

Transformer-based: BIT [26], ChangeFormer [27], AMTNet [49], ChangeCLIP [28].

Hybrid CNN-transformer: DMATNet [50], DARNet [51], SSANet [52], STransUNet [29], MSCANet [40], DMINet [53], GAS-Net [54], HATNet [30].

C. Experimental Detail

In this study, we implemented the proposed CFNet using the PyTorch framework. The experiments were conducted on a machine with Ubuntu 24.04 LTS as the operating system. The GPU used was an NVIDIA GeForce RTX 3090, while the CPU was an Intel(R) Xeon(R) Gold 6130H CPU @ 2.10 GHz with 64 cores. For data augmentation, we applied random flipping, scaling, translation, rotation, along with the addition of Gaussian blur and contrast adjustment. Regarding training specifics, we utilized a single NVIDIA GeForce RTX 3090 GPU throughout the experiments, setting the batch size to 32. The AdamW optimizer was employed to optimize the training process, with an initial learning rate of 0.0005, dynamically adjusted using a cosine annealing scheduler. During training, we continuously monitored the model's intersection over union (IoU) and $F1$ score on the validation set. Whenever there was an improvement in either IoU or $F1$ compared to previous iterations, we saved the current model weights and subsequently evaluated the model's performance on the test set.

D. Evaluation Metrics

To evaluate the effectiveness of our model, we focused on four key metrics: IoU, $F1$ score, recall, and precision. IoU is vital for assessing the accuracy of change detection by measuring the overlap between predicted and actual changes. Precision reduces false positives (FP), ensuring that the detected changes are relevant, while recall minimizes missed detections, capturing all relevant changes. The $F1$ score balances precision and recall, providing a comprehensive view of model performance, especially in scenarios with class imbalances. These metrics were selected for their ability to offer a detailed understanding of both overall and individual performance, crucial for assessing our algorithm's effectiveness. The calculation formulas are as follows:

$$\text{IoU} = \frac{\text{TP}}{\text{TP} + \text{FP} + \text{FN}} \quad (13)$$

$$\text{Recall} = \frac{\text{TP}}{\text{TP} + \text{FN}} \quad (14)$$

$$\text{Precision} = \frac{\text{TP}}{\text{TP} + \text{FP}} \quad (15)$$

$$F1 = 2 \cdot \frac{\text{Precision} \cdot \text{Recall}}{\text{Precision} + \text{Recall}} \quad (16)$$

True positives (TP) refer to correctly identified positive pixels, indicating that the model accurately detects changes. FP represent incorrectly identified positives, where the model mistakenly detects changes that are not present. False negatives (FN) are missed positives, meaning the model fails to detect actual changes, while true negatives (TN) denote correctly identified negatives, where the model accurately identifies areas without changes. Together, these terms are essential for evaluating model performance in classification tasks, particularly in change detection task.

TABLE II
QUANTITATIVE RESULTS ON CLCD

Model	<i>F1</i>	IoU	Rec	Prec
CDNet	65.15	48.31	59.43	72.09
STANet	67.97	51.49	64.16	72.26
P2V	70.22	54.10	65.93	75.11
MSCANet	71.65	55.83	67.07	76.91
HATNet	72.53	56.90	69.42	75.94
BIT	73.3	57.85	71.84	74.82
HCGMNet	74.67	59.58	73.92	75.44
DMINet	76.51	61.96	71.98	81.65
AMTNet	76.81	62.35	75.06	78.64
CGNet	77.05	62.67	71.71	83.25
EfficientCD	78.89	65.14	75.83	82.21
CFNet	81.41	68.65	81.08	81.74

The best results is highlight by using boldface.

TABLE III
QUANTITATIVE RESULTS ON LEVIR-CD

Model	<i>F1</i>	IoU	Rec	Prec
BIT	89.48	80.86	87.53	90.65
STANet	90.02	81.85	87.13	93.10
MSCANet	90.06	81.91	86.38	94.06
DDCNN	90.24	82.21	88.69	91.85
SNUNet	90.33	82.36	87.62	93.2
ChangeFormer	90.50	82.66	90.18	90.83
P2V	90.71	83.00	91.78	89.67
DMATNet	90.75	84.13	89.98	91.56
HATNet	91.38	84.12	90.15	92.64
STransUNet	91.41	84.19	90.55	92.30
ChangeCLIP	92.01	85.20	90.67	93.40
ChangerEx	92.06	85.29	90.56	93.61
CGNet	92.13	85.40	91.93	92.32
CFNet	92.18	85.49	90.86	93.54

The best results is highlight by using boldface.

TABLE IV
QUANTITATIVE RESULTS ON SYSU-CD

Model	<i>F1</i>	IoU	Rec	Prec
STANet	72.79	57.22	66.71	80.08
MSCANet	77.33	63.04	72.40	82.99
GAS-Net	78.50	64.61	78.22	78.77
BIT	78.65	64.81	73.68	84.34
SNUNet	79.54	66.02	75.87	83.58
P2V	79.73	66.29	79.29	80.17
HCGMNet	79.76	66.33	74.15	86.28
CGNet	79.92	66.55	74.37	86.37
HATNet	80.24	67.00	78.23	82.36
AFCF3D-Net	80.75	67.71	74.16	88.63
DARNet	81.03	68.10	79.11	83.04
SSANet	81.08	68.18	79.73	82.48
CDNeXt	81.36	68.57	74.10	90.20
CFNet	82.87	70.77	84.53	81.31

The best results is highlight by using boldface.

CLCD dataset, surpassing all classical change detection algorithms. Although the precision is lower than GaMPF's 84.6%, CFNet effectively achieves the tradeoff between recall and precision, exceeding GaMPF by 12.72% in recall. Notably, CFNet demonstrates significant improvements in IoU and *F1* scores compared to other leading change detection algorithms, with an *F1* score increase of 2.52% and an IoU increase of 3.51% over EfficientCD. Furthermore, as shown in Table III, CFNet achieves an *F1* score of 92.18% and an IoU of 85.49% on the LEVIR-CD dataset, with a slight improvement of 0.05% in *F1* and 0.09% in IoU compared to CGNet. As depicted in Table IV, on the SYSU-CD dataset, CFNet achieves an *F1* score of 82.87%, an IoU of 70.77%, and a recall of 84.53%, marking increases of 1.07% in *F1*, 2.59% in IoU, and 4.8% in recall compared to SSANet. The outstanding performance of CFNet across these three datasets clearly illustrates its effectiveness and robustness in RSCD task.

In Figs. 8–10, the visualization results of CFNet on the CLCD, LEVIR-CD, and SYSU-CD datasets are presented. For each dataset, the visualization includes Image T1, Image T2, ground truth, and the predictions made by CFNet, along with the predictions from several top-performing algorithms for comparative analysis. From Figs. 5–8, it is evident that CFNet demonstrates superior performance across all three datasets compared to other change detection algorithms. In addition, in the visualization results, we denote TP in white, TN in black, FN in red, and FP in green.

F. Ablation Study

To further validate the effectiveness of the proposed focuser module and content-aware strategy in RSCD task, we conducted ablation experiments for both content-aware and focuser. The

E. Qualitative Analysis and Visualization

As shown in Tables II–IV, we validated the proposed CFNet on three well-known public RSCD datasets: CLCD, LEVIR-CD, and SYSU-CD. We compared the performance metrics of our experimental results with several state-of-the-art algorithms in RSCD, demonstrating that CFNet achieves state-of-the-art levels for each dataset.

As indicated in Table II, CFNet achieves an *F1* score of 81.41%, an IoU of 68.65%, and a recall of 81.08% on the

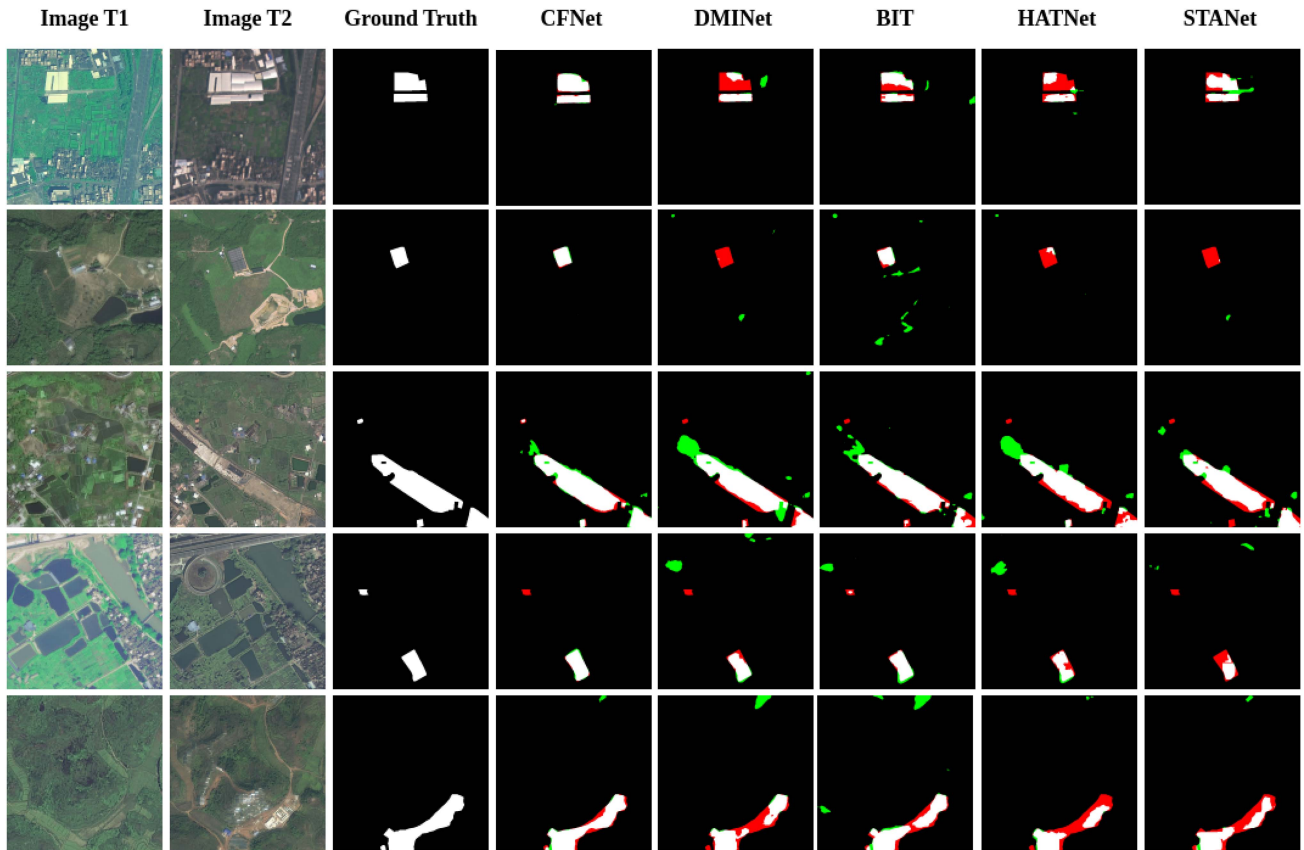


Fig. 8. Visualization result in CLCD dataset.

TABLE V
ABLATION STUDY IN IOU INDEX

Model	Content-Aware	Focuser	CLCD	SYSU-CD	LEVIR-CD
CFNet	×	×	62.76	68.25	84.19
	✓	×	63.85	68.36	84.54
	×	✓	64.65	68.94	84.90
	✓	✓	68.65	70.77	85.49

The best results is highlight by using boldface.

results for IoU and $F1$ scores on the CLCD, SYSU-CD, and LEVIR-CD datasets are presented in Tables V and VI, respectively. In the model where both content-aware and focuser are ablated, the outputs from the two content decoder modules are directly fed into the change decoder module, and L_{main} serves as the final loss function. In the model with the focuser module ablated, we retained the output of focuser module for “change content collection” and “unchanged content collection” to ensure the integrity of content-aware; however, the outputs $RM_i, i = 1, 2, 3, 4$ from the focuser module were removed, preventing the model from focusing progressively on changed areas within the change decoder module. In the model ablated for content-aware, we eliminated the processes for computing “change content collection” and “unchanged content collection,”

while using L_{main} as the final loss function. The results from the ablation experiments indicate that both focuser and content-aware effectively enhance the model’s performance in change detection task. Moreover, they exhibit a synergistic effect; when both modules are present, the model’s performance is further enhanced.

To analyze the impact of the loss weight ratio on the model performance, we conducted another ablation study by varying the ratio of α to β (where β and γ are set to the same value to maintain a consistent ratio between L_{cc} and L_{ucc}). Table VII presents the results of this study. Our original model setting corresponds to a ratio of 10 : 1. In our original model, we set $\alpha = 1$ and $\beta = \gamma = 0.1$, corresponding to the 10 : 1 ratio in the table. This choice was based on observing the absolute

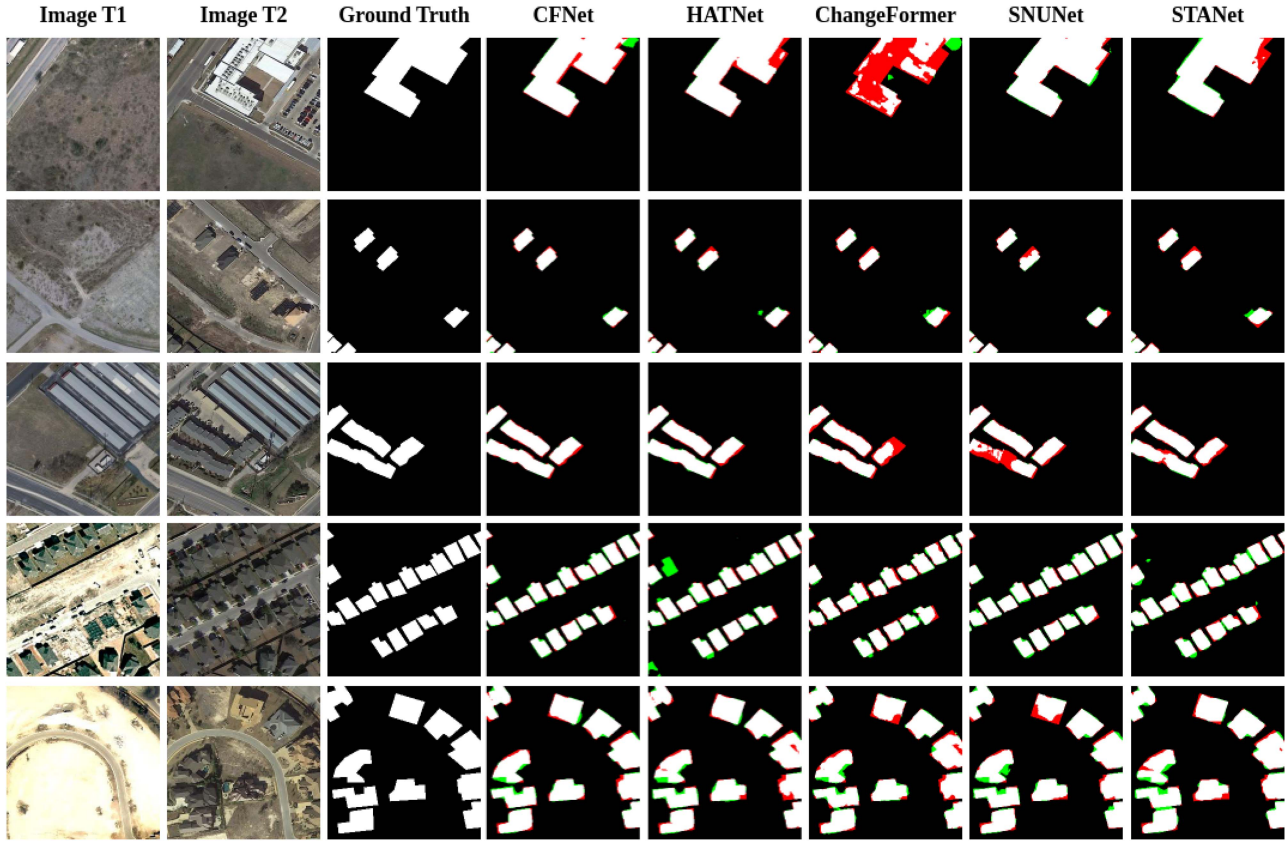


Fig. 9. Visualization result in LEVIR-CD dataset.

TABLE VI
ABLATION STUDY IN $F1$ INDEX

Model	Content-Aware	Focuser	CLCD	SYSU-CD	LEVIR-CD
CFNet	×	×	77.12	81.13	91.42
	✓	×	77.94	81.21	91.62
	×	✓	78.53	81.62	91.83
	✓	✓	81.41	82.89	92.18

The best results is highlight by using boldface.

TABLE VII
ABLATION STUDY ON THE $\alpha:\beta(\gamma)$ RATIO

$\alpha:\beta(\gamma)$	IoU	$F1$	Rec	Prec
100:1	62.64	77.78	76.87	78.72
50:1	63.84	78.00	79.62	76.31
10:1	68.65	81.41	81.08	81.74
1:1	60.88	75.68	74.71	76.69
1:5	58.92	74.15	74.00	74.29
1:10	56.24	71.99	71.64	72.35

The best results is highlight by using boldface.

magnitudes of L_{main} , L_{cc} , and L_{ucc} , ensuring they remain on a similar scale to prevent any single component from excessively dominating the model's fitting process. From Table VII, we observe that increasing the ratio of α results in L_{main} dominating the model, thereby reducing the influence of L_{cc} and L_{ucc} . This leads to a decline in model performance. Conversely, when increasing the ratio of β and γ , the auxiliary losses (L_{cc} and L_{ucc}) overly influence the fitting process, preventing the model from effectively learning the ground truth information through L_{main} . This causes the model's performance to degrade even more rapidly. The results indicate that a well-balanced loss ratio is crucial for achieving optimal performance.

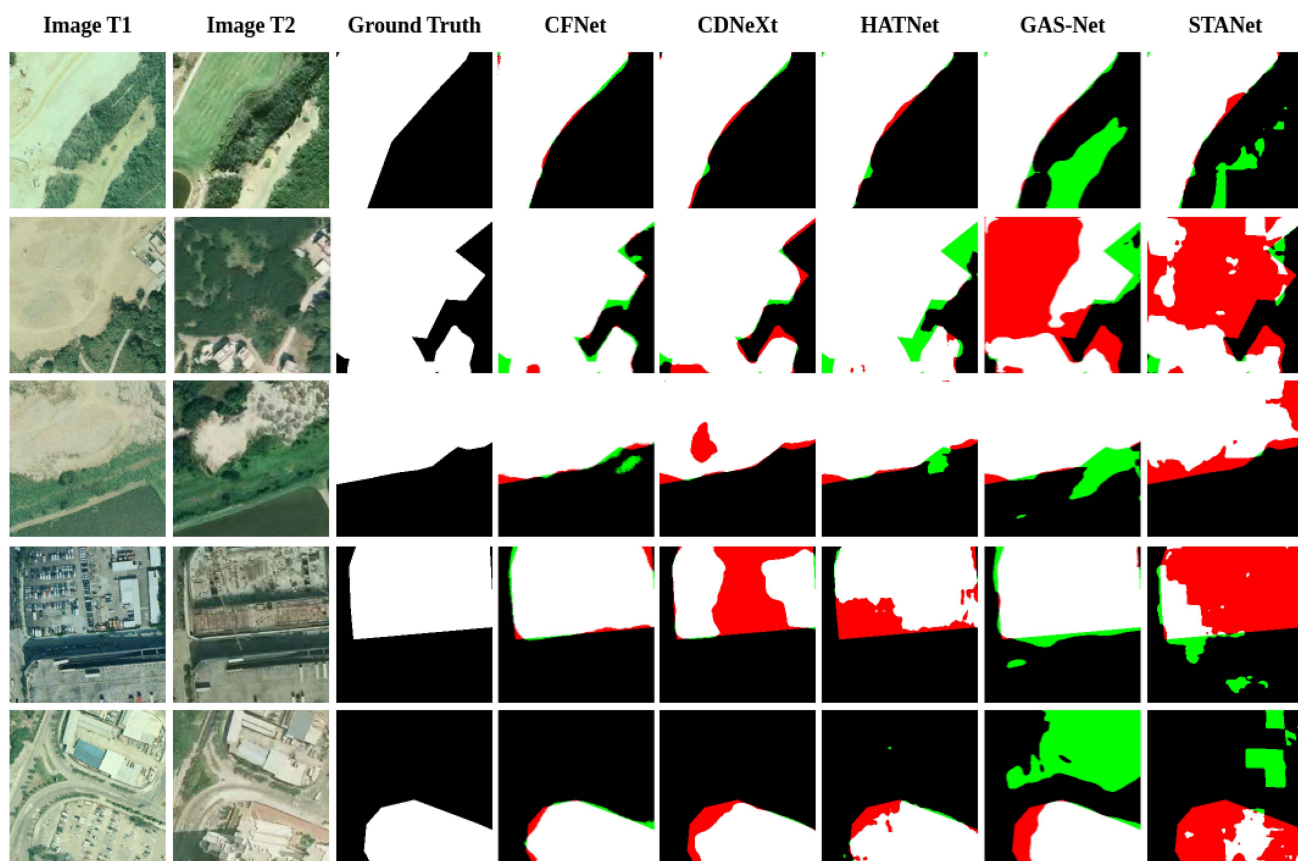


Fig. 10. visualization result in SYSU-CD dataset.

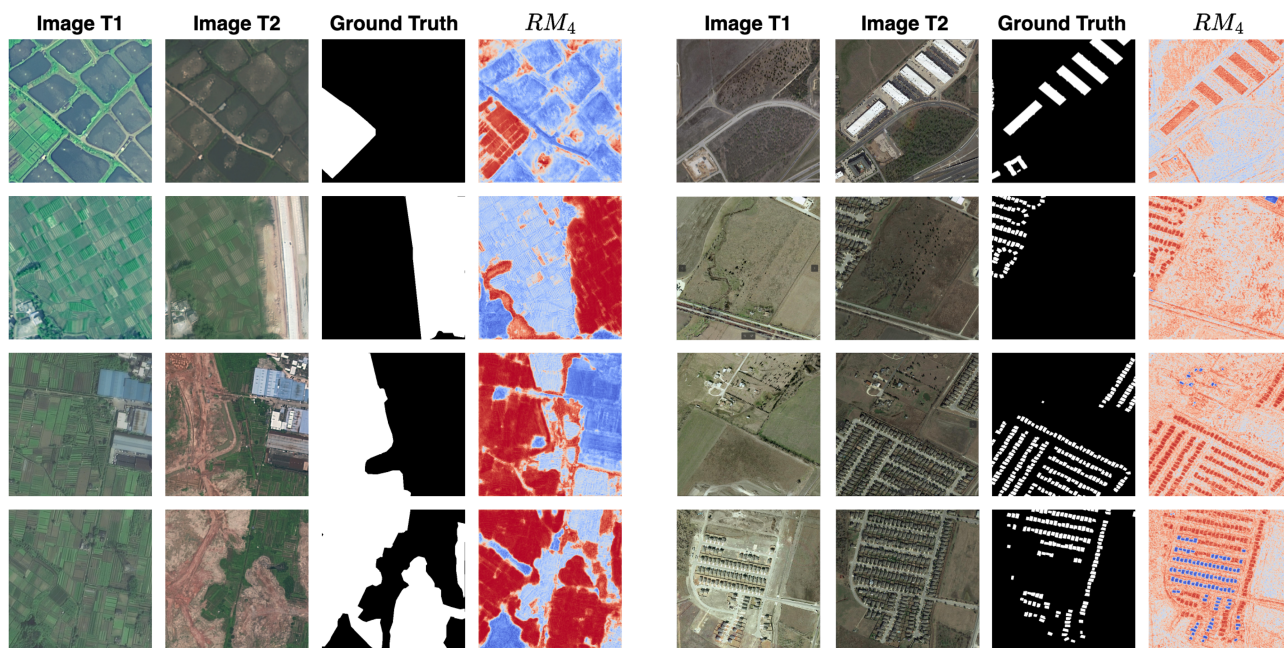


Fig. 11. Visualization of RM_4 from the test dataset of CLCD and LEVIR-CD. The left half demonstrates results on CLCD and the right half demonstrates results on LEVIR-CD. In the visualization of RM_4 , pixels with values closer to 1 are represented in red, indicating a higher likelihood of being a changed area, while pixels with values closer to 0 are represented in blue, indicating a higher likelihood of being an unchanged area.

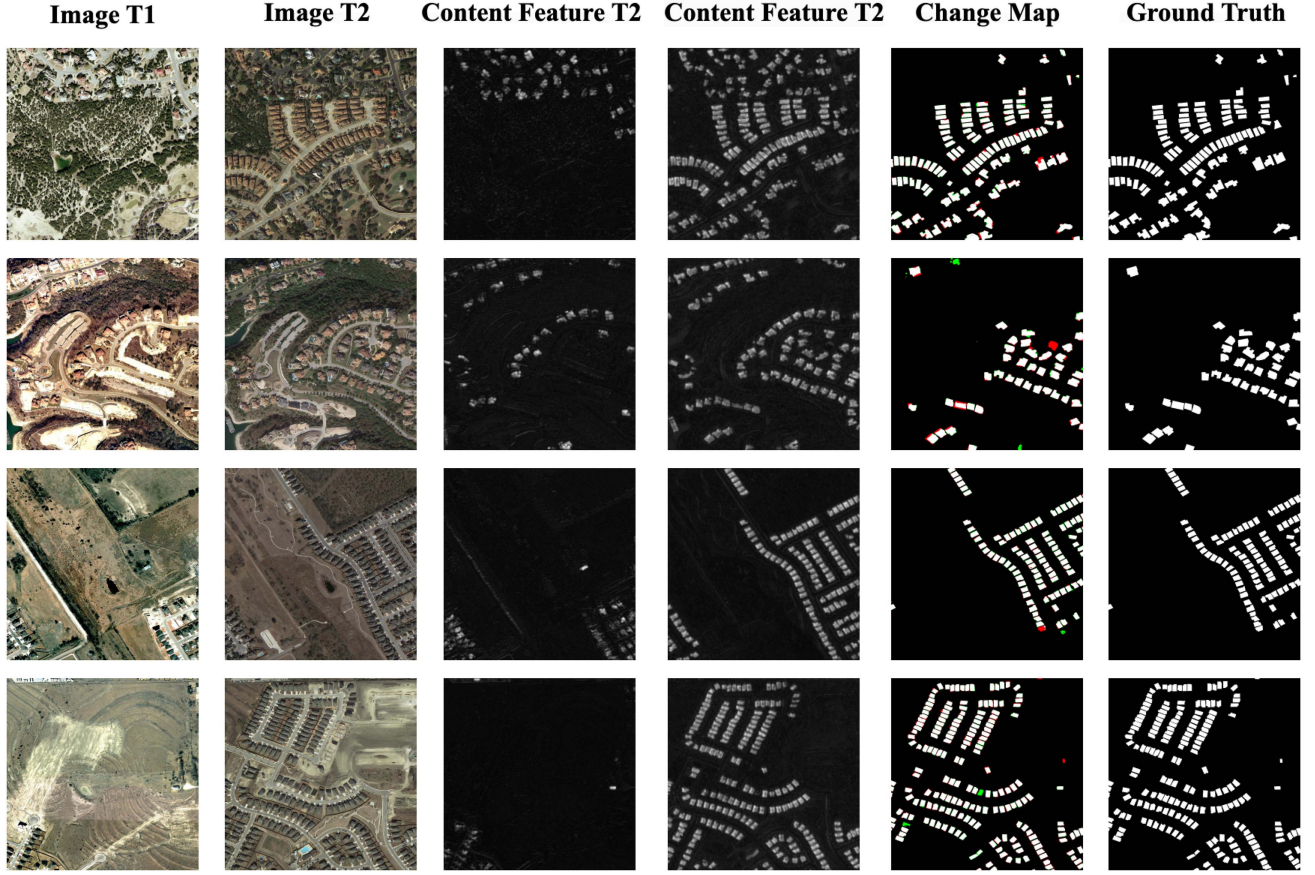


Fig. 12. Visualization of the content decoder’s output on the LEVIR-CD dataset. Each row presents a bitemporal image pair (first and second columns), the corresponding grayscale feature maps output by the content decoder (third and fourth columns), the final predicted “change map” and the ground truth. notably, since the content decoder is also influenced by L_{main} and the LEVIR-CD dataset primarily focuses on building changes, the content features predominantly capture structural information related to buildings.

V. DISCUSSION

A. Focuser Module in CFNet

In this study, we employed the focuser module to separately generate the “change content collection” and the “unchanged content collection.” the focuser module plays a critical role in our framework by distinguishing between areas that exhibit changes and those that remain constant, which is essential for the effectiveness of the change detection task. A pivotal step in the focuser module involves the computation of RM_i , where $i = 1, 2, 3, 4$. Among these, RM_4 is particularly noteworthy due to its proximity to the final change map within the network architecture. As the last stage in this process, RM_4 holds the most significant influence on the model’s performance in identifying changed areas accurately.

To evaluate the effectiveness of the focuser module in enhancing the change detection capability, we conducted an extensive visualization analysis of RM_4 . This visualization is critical as it provides direct insights into how well the focuser module aligns with the ground truth and contributes to the overall learning process of the network. Specifically, we selected RM_4 for analysis because it encapsulates the refined features and predictions generated by preceding modules, thus serving as the

most representative output for assessing the performance of the focuser module. The results of this visualization are depicted in Fig. 11, where we present the outputs of RM_4 derived from the test dataset of CLCD and LEVIR-CD. In these visualizations, pixels are assigned values within the range of $[0, 1]$, where higher values, closer to 1, are rendered in red to signify a greater likelihood of corresponding to changed areas. Conversely, pixels with lower values, closer to 0, are shown in blue, indicating a higher probability of being part of unchanged areas. This intuitive color mapping enables a clear interpretation of the model’s predictions.

The visualization results of RM_4 demonstrate a remarkable alignment with the ground truth, highlighting the efficacy of the focuser module in accurately distinguishing between changed and unchanged areas. The ability of RM_4 to closely approximate the Ground Truth underscores the module’s capacity to extract and refine discriminative features relevant to the change detection task. Furthermore, the clear separation between changed and unchanged areas in the visualizations validates the module’s robustness in enhancing the overall network’s ability to fit the ground truth, thereby improving the precision of the change detection outcomes.

B. Content-Aware Strategy in CFNet

Traditional change detection algorithms often overlook the impact of style differences between bitemporal images on model performance, especially given the strong fitting capabilities of DNN. In this study, inspired by self-similarity and the human ability to recognize content through internal structural features, we propose the content-aware strategy. In addition to the main branch where the model fits the ground truth, we introduce two auxiliary branches that decode content features. These branches leverage the previously mentioned L_{cc} and L_{ucc} to guide the model in learning content features for both changed and unchanged areas, thereby imposing stronger constraints on the main branch's fitting of the changed areas.

To further illustrate the effectiveness of the content-aware strategy in extracting content features, we visualize the largest-scale feature maps output by the content decoder. As direct visualization of all feature channels is not feasible, we compute the channel-wise mean and normalize the values to the [0,255] range to generate grayscale images. Despite the inevitable loss of some feature details, the resulting images provide an intuitive understanding of how the content decoder captures content information while filtering out style differences.

Fig. 12 presents four randomly selected cases from the LEVIR-CD dataset. In each row, the first and second columns are the input bitemporal images, which exhibit noticeable style variations. The third and fourth columns show the corresponding grayscale representations of the largest-scale feature maps output by the content decoder. These feature maps effectively suppress style differences while preserving structural content, highlighting the content decoder's ability to output content features. (It is important to note that the content decoder is also influenced by L_{main} . Since the LEVIR-CD dataset primarily focuses on building changes, the content features predominantly represent structural information related to buildings.) This ensures that the subsequent change decoder operates on more consistent content representations, thereby improving the accuracy of change detection.

However, in this study, the ratio of α and β , which defines the overall loss, was manually determined by observing the behavior of L_{main} , L_{cc} , and L_{ucc} during training. Therefore, it remains an open question how to set this ratio more scientifically, or even enable the model to flexibly adjust the ratio at different stages of training. I believe that a more rigorous and adaptable adjustment of α and β will further enhance the performance of CFNet in remote sensing image change detection task.

VI. CONCLUSION

In this article, we presented CFNet, a novel framework for RSCD that addresses the challenges posed by style variations between bitemporal images. By focusing on content features, CFNet reduces the impact of unpredictable style differences, which often hinder the accuracy of change detection models. The introduction of the content-aware strategy enhances the model's ability to capture intrinsic content features, while the focuser module allows dynamic emphasis on both changed and unchanged areas throughout the detection process.

Our extensive experiments on CLCD, LEVIR-CD, and SYSU-CD demonstrate that CFNet consistently outperforms existing state-of-the-art methods. Notably, CFNet achieved significant improvements in $F1$ score and IoU, illustrating its robustness and generalizability across diverse remote sensing scenarios. Furthermore, our ablation studies highlight the complementary roles of the content-aware strategy and the focuser module in enhancing detection accuracy.

Future work could explore more adaptive mechanisms for balancing the loss components of CFNet, potentially enabling further improvements in model performance. Moreover, expanding the model's applicability to other remote sensing domains, such as environmental monitoring and disaster response, represents a promising direction for future research.

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